PQC DAILY STUDY SERIES

# Deep Dive: Understanding CRYSTALS-Dilithium (ML-DSA)

**Day 40** | Category: PQC Foundational Concepts Date: 2026-10-25 | DAILY MONITORING

## Deep Dive: Understanding CRYSTALS-Dilithium (ML-DSA)

Following our exploration of ML-KEM (Kyber), today we delve into its counterpart for digital signatures: **CRYSTALS-Dilithium**, standardized by NIST as the **Module-Lattice-based Digital Signature Algorithm (ML-DSA)** in FIPS 204. While ML-KEM ensures confidentiality, ML-DSA guarantees authenticity and integrity, forming a complete suite for post-quantum secure communications.

ML-DSA is designed to be a general-purpose digital signature scheme that is secure against attacks from both classical and quantum computers. It is based on the hardness of finding short vectors in module lattices, a problem believed to be computationally difficult even for quantum algorithms.

### Applicability to Internet Protocols & Cybersecurity

Digital signatures are fundamental to internet security, used to authenticate software, messages, and parties in a communication. ML-DSA is poised to replace classical signature schemes like RSA and ECDSA in numerous protocols:

- **TLS Certificates:** Certificate Authorities (CAs) will begin issuing X.509 certificates with ML-DSA signatures. This ensures that when a browser connects to a server, it can verify the server's identity in a quantum-resistant manner.
- **Code Signing:** To prevent the distribution of malicious software, developers sign their code. Using ML-DSA for code signing protects the software supply chain from quantum-era threats, ensuring that firmware, software updates, and applications are authentic.
- **Secure Boot & Firmware:** In the boot process of a device, each component's signature is checked to ensure it hasn't been tampered with. ML-DSA provides the necessary security to protect the integrity of device firmware (e.g., in the Linux Kernel, IoT devices, etc.) against future attacks.
- **JSON Web Tokens (JWTs):** ML-DSA can be used to sign JWTs, providing a quantum-resistant method for securing stateless authentication tokens in web applications.

### Hybrid Signatures: A Phased Transition

Similar to hybrid key exchange, protocols will likely adopt a hybrid signature approach initially. This involves using both a classical signature (like ECDSA) and a post-quantum signature (ML-DSA) simultaneously. This strategy ensures security against both classical and quantum adversaries during the transition period, providing backward compatibility and mitigating risks from potential weaknesses in new algorithms.

## Key Metrics & Comparison

Let's compare ML-DSA with the classical ECDSA and another PQC signature standard, SLH-DSA (SPHINCS+).

| Metric | ECDSA (P-256) | ML-DSA-65 (Security Level 3) | SLH-DSA-128f (Security Level 1) |
|---|---|---|---|
| **Underlying Problem** | Elliptic Curve Discrete Logarithm | Module-LWE/SIS | Security of Hash Functions |
| **Public Key Size** | ~64 bytes | ~1952 bytes | ~32 bytes |
| **Signature Size** | ~70 bytes | ~3291 bytes | ~7856 bytes |
| **Performance** | Fast | Very Fast (fastest lattice-based) | Slow Signing, Fast Verification |
| **Key Generation** | Fast | Fast | Slow |
| **Statefulness** | Stateless | Stateless | Stateful (private key must be managed) |

ML-DSA offers a strong balance of performance and security, though its key and signature sizes are significantly larger than ECDSA. Its stateless nature makes it a more direct replacement for classical schemes compared to the stateful SLH-DSA.

## Daily Quiz

**1. What is the primary role of ML-DSA in post-quantum cryptography?**

- ○ To encrypt communications.
- ○ To establish a shared secret.
- ○ To provide authenticity and integrity through digital signatures.

**2. In which of the following applications is ML-DSA NOT a suitable replacement for classical signatures?**

- ○ TLS Certificates.
- ○ Key Encapsulation Mechanisms (KEMs).
- ○ Code Signing.

**3. What is a key advantage of ML-DSA over SLH-DSA (SPHINCS+)?**

- ○ ML-DSA has smaller signature sizes.
- ○ ML-DSA is stateless, making it easier to deploy.
- ○ ML-DSA relies on hash functions, which are better understood.

---